

Computational Astrophysics Exercises — Assignments Set 6

Numerical treatment of ordinary differential equations — Cont'd

Exercise 1: The Restricted Circular Three Body Problem

We have not covered the RC3BP in the lecture, but you find more than a good introduction in the book by Murray and Dermott¹. To work on this exercise, you do not need to read the whole chapter, and can focus on the numerical implementation and plotting fun.

The equations of motion of a massless body in the restricted three-body problem in the corotating frame are given in dimensionless units by

$$\ddot{x} - 2\Omega\dot{y} - \Omega^2 x = - \left[\mu_1 \frac{x + \mu_2}{r_1^3} + \mu_2 \frac{x - \mu_1}{r_2^3} \right], \quad (1)$$

$$\ddot{y} + 2\Omega\dot{x} - \Omega^2 y = - \left[\frac{\mu_1}{r_1^3} + \frac{\mu_2}{r_2^3} \right] y, \quad (2)$$

where $\mu_1 = m_1/(m_1 + m_2)$, $\mu_2 = m_2/(m_1 + m_2)$ are the reduced masses of the two bodies with distance $a = 1$. The distances r_1 and r_2 to the two masses μ_1 and μ_2 are given by $r_1^2 = (x + \mu_2)^2 + y^2$ and $r_2^2 = (x - \mu_1)^2 + y^2$ using the normalisation $\mu_1 + \mu_2 = 1$. The geometry of the problem is shown in Figure 1. In the dimensionless units, the angular velocity is $\Omega = 2\pi/P = 1$. Hence, one period of the two masses μ_1 and μ_2 around each other is exactly 2π .

The only known conserved quantity for the circular restricted three-body problem is the Jacobi integral or Jacobi constant

$$C_J = \Omega^2(x^2 + y^2) + 2 \left(\frac{\mu_1}{r_1} + \frac{\mu_2}{r_2} \right) - (\dot{x}^2 + \dot{y}^2). \quad (3)$$

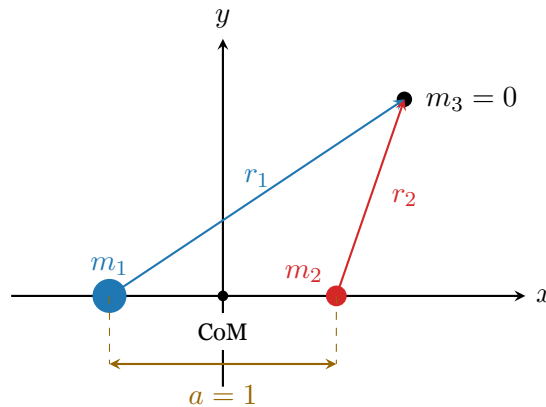


Figure 1: The geometry of the problem. The center of mass is in the origin.

¹You can access it via the University of Tübingen, or see the chapter on ILLAS in the assignment sets folder.

Exercises

- [1. Bonus points: Proof that the Jacobi constant C_J given by equation (3) is indeed a conserved quantity.]
2. Write a program to solve numerically the equations of motion (1) and (2). If possible use the classical RK4 integrator, or at least the Heun scheme. Please use the following initial conditions for each numerical integration: $\dot{x}_0 = 0$ and $y_0 = 0$. The input parameters of your program will be C_J and x_0 . With these values, you can then calculate the initial velocity $\dot{y}_0$ using equation (3). Note: Equation (3) only yields $\dot{y}_0^2$. Choose $\dot{y}_0 > 0$, meaning the test particle starts with a velocity in the positive y -direction. Now, choose $\mu_2 = 10^{-3}$ and $C_J = 3.03$ and integrate the equations of motion for different initial positions x_0 . Choose a constant timestep of $\Delta t = 10^{-3}$ and integrate for 5×10^6 steps.
We want to analyse the system for different initial positions $x_0 \in [0.2, 0.8]$. Choose the following values for $x_0 = (0.21, 0.24, 0.26, 0.27, 0.4, 0.5, 0.6, 0.8)$. The questions are: which values of x_0 lead to periodic solutions, and which values are unstable. To answer these questions, use the following approach:
 - (a) Print out the positions x, y of the particle at certain timesteps (e.g., every 500 timesteps) and plot them. Do you find any symmetries?
 - (b) Create Poincaré sections using the following method: Print out the x -coordinate and the x -velocity each time when the particle crosses the y -axis in positive direction ($y = 0$ and $\dot{y} > 0$). Plot these values in a $\dot{x}$ - x diagram and analyse its structure, look for resonant isles or chaotic regions.

(10 (+2 bonus) points)

A note on the use of AI 🤖:

The use of AI tools (e.g. ChatGPT, Claude, Copilot) is permitted for these exercises, provided that (1) any use is clearly documented in your submission (which tool, for what purpose), and (2) you are able to explain all submitted solutions in full — on request, during the exercise class or otherwise. Submissions that cannot be explained by the submitter(s) (that includes all members of the submitting team) will be graded as failed, regardless of whether AI was used.

in total

(0b1010 (+0b10 bonus) points)

Deadline:

Fri, 19 June at noon 🕒 via email (including code) to `christoph.schaefer@uni-tuebingen.de`.